Build and Configure a Small Office Network

Q1. Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.

Solution : Configure five computers in the same subnet and verify connectivity using ping.

IP Address Table

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.10.11	255.255.255.0	192.168.10.1
PC1	192.168.10.12	255.255.255.0	192.168.10.1
PC2	192.168.10.13	255.255.255.0	192.168.10.1
PC3	192.168.10.14	255.255.255.0	192.168.10.1
PC4	192.168.10.15	255.255.255.0	192.168.10.1

Steps

1. Open Cisco Packet Tracer and add five PCs and one switch.
2. Connect each PC to the switch using Copper Straight-Through cables.
3. On each PC, open Desktop → IP Configuration.
4. Enter the IP address, subnet mask, and gateway from the table.
5. Open PC0's Command Prompt and run:

ping 192.168.10.12

ping 192.168.10.13

ping 192.168.10.14

ping 192.168.10.15

Q2. Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.

Solution : Automatically assign unique IP addresses to three or more devices.

DHCP Configuration

Add one server, one switch, and three PCs in Packet Tracer. Connect them using Ethernet cables.

Configure the server's static IP address:

Setting	**Value**
Server IP	192.168.10.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1

Go to Server → Services → DHCP, enable DHCP, and configure:

Setting	**Value**
Pool Name	OfficePool
Default Gateway	192.168.10.1
DNS Server	192.168.10.2
Start IP Address	192.168.10.100
Subnet Mask	255.255.255.0
Maximum Users	50

Steps

1. Enable the DHCP service.
2. Save the configuration.
3. Open each PC's Desktop → IP Configuration.
4. Select DHCP.
5. Check the assigned addresses using ipconfig.

Q3. Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface. Hint : You can map hostnames to local IPs using the DNS service or by editing the hosts file for a quick test.

Solution : Configure local name resolution so that a hostname points to a local web server.

Steps

1. Add a server to Packet Tracer and configure its IP address as 192.168.10.2.
2. Open Services → DNS.
3. Turn the DNS service On.
4. Add this DNS record:

Name	Type	Address
musicplayer.local	A	192.168.10.2

5. Ensure the client PCs use 192.168.10.2 as their DNS server, either through DHCP or manual configuration.
6. Enable HTTP under the server's Services menu.
7. Open a PC's web browser and enter : http://musicplayer.local

Q4. Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN. Hint : Use ping or file sharing to test connectivity between devices in different VLANs.

Solution : Divide the network into two logical segments and test communication between devices.

VLAN Plan

VLAN	Name	Devices	Subnet
VLAN 10	Workstations	PC0, PC1	192.168.10.0/24
VLAN 20	Smart Devices	PC2, PC3	192.168.20.0/24

Example Switch Configuration

Open the switch CLI and enter:

enable

configure terminal

vlan 10

name Workstations

exit

vlan 20

name Smart_Devices

exit

interface range fa0/1-2

switchport mode access

switchport access vlan 10

exit

interface range fa0/3-4

switchport mode access

switchport access vlan 20

exit

end

show vlan brief

Connect PC0 and PC1 to switch ports Fa0/1 and Fa0/2. Connect PC2 and PC3 to Fa0/3 and Fa0/4.

Assign IP addresses:

Device	**VLAN**	**IP Address**
PC0	10	192.168.10.11
PC1	10	192.168.10.12
PC2	20	192.168.20.11
PC3	20	192.168.20.12

Q5. Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.

Solution : Configure a router to translate private LAN addresses into an address used on the external network.

Device/Interface	IP Address
Router LAN	192.168.10.1/24
PC0	192.168.10.11/24
PC1	192.168.10.12/24
Router WAN	203.0.113.2/24
External Server	203.0.113.10/24

The external network here is a simulated network, using example addresses for documentation.

Example Cisco Router Configuration

Assume GigabitEthernet0/0 connects to the LAN and GigabitEthernet0/1 connects to the external network.

enable

configure terminal

access-list 1 permit 192.168.10.0 0.0.0.255

interface gigabitEthernet0/0

ip address 192.168.10.1 255.255.255.0

ip nat inside

no shutdown

exit

interface gigabitEthernet0/1

ip address 203.0.113.2 255.255.255.0

ip nat outside

no shutdown

exit

ip nat inside source list 1 interface gigabitEthernet0/1 overload

end

For external connectivity, also configure the appropriate routing and the external server's gateway. The router interfaces and IP addresses must match your actual Packet Tracer topology.

Testing

From PC0, run:

ping 203.0.113.10

On the router, check NAT translations:

show ip nat translations

show ip nat statistics